

renal failure, coma

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration of oral formulation (at least 4 weeks when possible), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after loxapine is discontinued

Pharmacokinetics

- Half-life approximately 4 hours for oral formulation
- Half-life approximately 12 hours for intramuscular formulation
- Multiple active metabolites with longer half-lives than parent drug
- * *N*-desmethyl loxapine is amoxapine, an antidepressant

- 8-hydroxyloxapine and 7-hydroxyloxapine are also serotonin-dopamine antagonists
- 8-hydroxyamoxapine and 7-hydroxyamoxapine are also serotonin-dopamine antagonists



Drug Interactions

- Respiratory depression may occur when loxapine is combined with lorazepam
- Additive effects may occur if used with CNS depressants
- May decrease the effects of levodopa, dopamine agonists
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome
- Combined use with epinephrine may lower blood pressure
- May increase the effects of antihypertensive drugs except for guanethidine, whose antihypertensive actions loxapine may antagonize



Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold

- Antiemetic effect can mask signs of other disorders or overdose
- Do not use epinephrine in event of overdose, as interaction with some pressor agents may lower blood pressure
- Use cautiously in patients with glaucoma, urinary retention
- Observe for signs of ocular toxicity (pigmentary retinopathy, lenticular pigmentation)
- Avoid extreme heat exposure
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is in a comatose state or has CNS depression
- If patient has asthma or history of asthma, COPD, other lung disease associated with bronchospasm, acute respiratory signs/symptoms, current use of medications to treat airways, history of bronchospasm following treatment with loxapine inhalation powder (inhalant only)

- If there is a proven allergy to loxapine
- If there is a known sensitivity to any dibenzoxazepine

Special Populations

Renal Impairment

- Use with caution

Hepatic Impairment

- Use with caution

Cardiac Impairment

- Use with caution

Elderly

- Some patients may tolerate lower doses better
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

Children and Adolescents

- Safety and efficacy not established
- Generally, consider second-line after atypical antipsychotics

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Renal papillary abnormalities have been seen in rats during pregnancy
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

Breast Feeding

- Unknown if loxapine is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Intramuscular formulation for emergency use

Potential Disadvantages

- Patients with tardive dyskinesia

Primary Target Symptoms

- Positive symptoms of psychosis
- Motor and autonomic hyperactivity
- Violent or aggressive behavior

Pearls

- * Recently discovered to be a serotonin dopamine antagonist (binding studies and PET scans)
- * Active metabolites are also serotonin dopamine antagonists with longer half-lives than parent drug, thus possibly allowing once daily

treatment

* One active metabolite is an antidepressant (amoxapine, also known as *N*-desmethyl-loxapine)

- Theoretically, loxapine should have antidepressant actions, especially at high doses, but no controlled studies
- Theoretically, loxapine may have advantages for short-term use in some patients with psychotic depression
- Developed as a conventional antipsychotic; i.e., reduces positive symptoms, but causes drug-induced parkinsonism and prolactin elevations
- Lower risk of drug-induced parkinsonism than haloperidol in some studies, but not fixed-dose studies and no low-dose studies

* Causes less weight gain than other antipsychotics, both atypical and conventional, and may even be associated with weight loss

- No formal studies of negative symptoms, but some studies show superiority to conventional antipsychotics for emotional withdrawal and social competence
- Best use may be as low-cost augmentation agent to atypical antipsychotics

* Enhances efficacy in clozapine partial responders when given concomitantly with clozapine

- For previously stabilized patients with “breakthrough” agitation or incipient decompensation, “top up” the atypical antipsychotic with

as-needed intramuscular or oral single doses of loxapine

- An inhalation powder for acute agitation is available only through a restricted program under a Risk Evaluation and Mitigation Strategy (REMS) called the ADASUVE REMS
- Due to the risk of bronchospasm, loxapine inhalation powder should be administered only in an enrolled healthcare facility that has immediate onsite access to equipment and personnel trained to manage acute bronchospasm, including advanced airway management (intubation and mechanical ventilation)
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another
- Patients with inadequate responses to atypical antipsychotics may benefit from determination of plasma levels and, if low, a dosage increase even beyond the usual prescribing limits
- Patients with inadequate responses to atypical antipsychotics may also benefit from a trial of augmentation with a conventional antipsychotic such as loxapine or from switching to a conventional antipsychotic such as loxapine
- However, long-term polypharmacy with a combination of a conventional antipsychotic such as loxapine with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either

- For treatment-resistant patients, especially those with impulsivity, aggression, violence, and self-harm, long-term polypharmacy with 2 atypical antipsychotics or with 1 atypical antipsychotic and 1 conventional antipsychotic may be useful or even necessary while closely monitoring. In such cases, it may be beneficial to combine 1 depot antipsychotic with 1 oral antipsychotic
- Although a frequent practice by some prescribers, adding 2 conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

Suggested Reading

Chakrabarti A , Bagnall A , Chue P , et al. Loxapine for schizophrenia .
Cochrane Database Syst Rev 2007 ;**17** (4):CD001943 .

Fenton M , Murphy B , Wood J , et al. Loxapine for schizophrenia .
Cochrane Database Syst Rev 2000 ;(2):CD001943 .

Heel RC , Brogden RN , Speight TM , Avery GS . Loxapine: a review of its pharmacological properties and therapeutic efficacy as an antipsychotic agent . *Drugs* 1978 ;**15** (3):198 –217 .

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939-51 .

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Zisook S , Click MA Jr . Evaluations of loxapine succinate in the
ambulatory treatment of acute schizophrenic episodes . *Int*
Pharmacopsychiatry 1980 ;**15** (6):365–78 .

Lumateperone

Therapeutics

Brands

- Caplyta
- *see index for additional brand names*

Generic?

- No

Class

- Atypical antipsychotic (serotonin-dopamine antagonist; second-generation antipsychotic; also a mood stabilizer)

Commonly Prescribed for

- *(bold for FDA approved)*
- **Schizophrenia**
- Bipolar depression
- Acute mania/mixed mania
- Bipolar maintenance
- Other psychotic disorders
- Treatment-resistant depression

- Behavioral disturbances in dementias
- Behavioral disturbances in children and adolescents
- Disorders associated with problems with impulse control
- Posttraumatic stress disorder

How the Drug Works

- Blocks serotonin 2A receptors, causing enhancement of dopamine release in certain brain regions and thus reducing motor side effects and possibly improving cognitive and affective symptoms
- Blocks postsynaptic dopamine 2 receptors, reducing positive symptoms of psychosis and stabilizing affective symptoms
- Binds as a partial agonist at presynaptic D2 receptors
- Interactions at a myriad of other neurotransmitter receptors may contribute to lurasidone's efficacy
- Binds to D1 receptors, with preclinical data showing D1-dependent activation of AMPA receptors leading to activation of the mTOR pathway, a mechanism that has been associated with antidepressant effects
- Also has affinity for the serotonin reuptake pump (SERT, serotonin transporter)

How Long Until It Works

- Psychotic and manic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior as well as on cognition and affective stabilization
- Classically recommended to wait at least 4–6 weeks to determine efficacy of drug, but in practice some patients require up to 16–20 weeks to show a good response, especially on negative or cognitive symptoms

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Can improve negative symptoms, as well as aggressive, cognitive, and affective symptoms in schizophrenia
- Most schizophrenia patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Perhaps 5–15% of schizophrenia patients can experience an overall improvement of greater than 50–60%, especially when receiving stable treatment for more than a year
- Such patients are considered super-responders or “awakeners” since they may be well enough to be employed, live independently, and sustain long-term relationships
- Continue treatment until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis

- For second and subsequent episodes of psychosis, treatment may need to be indefinite
- Even for first episodes of psychosis, it may be preferable to continue treatment indefinitely to avoid subsequent episodes
- Many bipolar patients may experience a reduction of symptoms by half or more
- Treatment may not only reduce mania but also prevent recurrences of mania in bipolar disorder

If It Doesn't Work

- Try one of the other atypical antipsychotics
- If 2 or more antipsychotic monotherapies do not work, consider clozapine
- Some patients may require treatment with a conventional antipsychotic
- If no first-line atypical antipsychotic is effective, generally do not consider higher doses of lurasidone, as higher doses do not have established efficacy
- May consider augmentation with lamotrigine; do not consider augmentation with valproate as this increases lurasidone levels
- Consider noncompliance and switch to another antipsychotic with fewer side effects, or to an antipsychotic that can be given by depot injection

- Consider initiating rehabilitation and psychotherapy such as cognitive remediation
- Consider presence of concomitant drug abuse

Best Augmenting Combos for Partial Response or Treatment Resistance

- Lamotrigine
- Lithium
- Topiramate
- Benzodiazepines

Tests

Before starting any atypical antipsychotic

- Weigh all patients and track BMI during treatment
- Get baseline personal and family history of diabetes, obesity, dyslipidemia, hypertension, and cardiovascular disease
- Get waist circumference (at umbilicus), blood pressure, fasting plasma glucose, and fasting lipid profile
- Determine if the patient
 - is overweight (BMI 25.0–29.9)
 - is obese (BMI >30)
 - has pre-diabetes (fasting plasma glucose 100–125 mg/dL)

- has diabetes (fasting plasma glucose >126 mg/dL)
- has hypertension (BP >140/90 mmHg)
- has dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL cholesterol)
- Treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

Monitoring after starting an atypical antipsychotic

- BMI monthly for 3 months, then quarterly
- Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- Blood pressure, fasting plasma glucose, fasting lipids within 3 months and then annually, but earlier and more frequently for patients with diabetes or who have gained >5% of initial weight
- Treat or refer for treatment and consider switching to another atypical antipsychotic for patients who become overweight, obese, pre-diabetic, diabetic, hypertensive, or dyslipidemic while receiving an atypical antipsychotic
- Even in patients without known diabetes, be vigilant for the rare but life-threatening onset of diabetic ketoacidosis, which always requires immediate treatment, by monitoring for the rapid onset of polyuria, polydipsia, weight loss, nausea, vomiting, dehydration, rapid respiration, weakness, clouding of sensorium, even coma

- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and lurasidone should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking serotonin 2A receptors, it can cause sedation
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- Sedation, dry mouth, nausea
- Tardive dyskinesia (TD) (reduced risk compared to conventional antipsychotics)
- Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration



Life-Threatening or Dangerous Side Effects

- Hyperglycemia, in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients taking atypical antipsychotics
- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Occurs less frequently with evening administration

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take in the evening to help reduce daytime sedation
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain
- Switch to another atypical antipsychotic

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism

- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 42 mg once daily

Dosage Forms

- Capsule 42 mg

How to Dose

- Initial dose 42 mg once daily; titration is not required
- Recommended to be taken with food because this is how clinical trials were performed; however, food increases the AUC by only 9%



Dosing Tips

- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly

- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as topiramate or lamotrigine
- Take in the evening to avoid daytime sedation
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Limited experience

Long-Term Use

- Studied in clinical trials for up to 1 year
- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- See Switching section of individual agents for how to stop lumateperone
- Rapid oral discontinuation may lead to rebound psychosis and worsening of symptoms

Pharmacokinetics

- Terminal half-life approximately 18 hours after intravenous administration
- Metabolized by multiple enzymes, including CYP450 3A4 and CYP450 1A2
- Absorption is delayed in the presence of food; ingestion of a high-fat meal reduces C_{max} by 33% and increases mean AUC by 9%



Drug Interactions

- Avoid concomitant use with moderate or strong CYP450 3A4 inhibitors (e.g., fluvoxamine, nefazodone, foods such as grapefruit juice)
- Avoid concomitant use with CYP450 3A4 inducers (e.g., carbamazepine, phenytoin, St. John's wort)
- Avoid concomitant use with UGT (uridine 5'-diphosphoglucuronosyltransferase) inhibitors (e.g., valproate)
- Little potential to affect metabolism of drugs cleared by CYP450 enzymes
- May enhance the effects of antihypertensive drugs
- May antagonize levodopa, dopamine agonists



Other Warnings/Precautions

- Use with caution in patients with conditions that predispose to hypotension (dehydration, overheating)
- Atypical antipsychotics have the potential for cognitive and motor impairment, especially related to sedation
- Atypical antipsychotics can cause body temperature dysregulation; use caution in patients who may experience conditions that increase body temperature (e.g., strenuous exercise, saunas, extreme heat, dehydration, or concomitant anticholinergic medication)
- Dysphagia has been associated with antipsychotic use, and lumateperone should be used cautiously in patients at risk for aspiration pneumonia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy
- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If there is a proven allergy to lumateperone

Special Populations

Renal Impairment

- Dose adjustment not necessary

Hepatic Impairment

- Mild impairment (Child-Pugh score between 5 and 6): no dose adjustment necessary
- Not recommended for use in patients with moderate to severe impairment

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic

Children and Adolescents

- Safety and efficacy have not been established
- Children and adolescents using lumateperone may need to be monitored more often than adults

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In rats and rabbits, administration of lumateperone during organogenesis did not result in malformations at doses 2.4 and 9.7 times, respectively, the maximum recommended human dose (MRHD) of 42 mg
- In rats, administration of lumateperone during organogenesis through lactation resulted in lower pup survival at 4.9 times the MRHD, but not at 2.4 times the MRHD